
ME 640: Solar Power Utilization

Course Code	ME 640
Course Name	Solar Power Utilization
L-T-P-C	3-0-0-3
Prerequisites	Instructor Consent
Intended for	UG/M. Tech. /MS/PhD
Distribution	Specialized stream elective course for M. Tech. in Mechanical Engineering with specialization in Energy Systems, and elective course for other students
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I**

Solar radiation and modelling, solar collectors and types: flat plate, concentrating solar collectors, advanced collectors and solar concentrators, Sensitive coating. [14 Lectures]

- **Module II**

Solar water heating, solar cooking, solar drying, solar distillation and solar refrigeration, Active and passive heating and cooling of buildings. [14 Lectures]

- **Module III**

Solar thermal power generation, solar cells, Home lighting systems, Solar lanterns, Solar PV pumps, Solar energy storage options, Industrial heat systems, Solar thermal power generation and sterling engine, Solar economics. [14 Lectures]

Text Books

1. Robert Foster, Majid Ghassemi, Alma Cota, **Solar Energy: Renewable Energy and the Environment**, CRC Press, 2009..